Reverse Osmosis Skid

Control & Operation Specification

EQSP32 Industrial IoT Controller

Document	Detail
Title	Reverse Osmosis Skid - Control & Operation Specification
Revision	v0.3 (draft for review)
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Controller	Erqos EQSP32 (single unit, on-board I/O only)
Prepared by	Erqos Technologies
Status	Draft - reflects sign-off changes (0-300 PSI sensors, TDS 10-min stop)

1. Purpose & Scope

This document specifies the automatic control and operation of a single-pass reverse osmosis (RO) water-treatment skid driven by one Erqos EQSP32 controller using only its on-board I/O. It defines the I/O, sensor scaling, operating states, control sequence, process monitoring and protection, membrane-fouling detection, operator interface (run/stop and RGB status indicator) and fault handling.

Cloud connectivity, telemetry/MQTT, data logging, temperature compensation, PID control and a feed/booster pump are out of scope for this revision (see Section 12).

2. System Overview

Feed (brackish) water is gravity-fed from the supply tank(s) through a supply electro-valve and cartridge pre-filters to the suction of a fixed-speed high-pressure RO pump. The pump forces feed water across the RO membrane. Permeate (product water) flows to the clean-water tank; the concentrate (brine) leaves through a fixed flow restrictor that sets the membrane operating pressure and then to drain (waste).

A flush valve is plumbed in parallel with the flow restrictor on the brine line. With the flush valve closed the brine passes through the restrictor and the membrane is pressurised for production. With the flush valve open the brine bypasses the restrictor straight to drain, dropping the pressure and giving a high cross-flow forward-flush of the membrane surface (the pump keeps running during a flush).

Controlled actuators:

- RO Pump - ON/OFF via a 24 V contactor coil (single speed).
- Supply (inlet) valve - 24 V solenoid; opens feed to the pump.
- Flush (brine-bypass) valve - 24 V solenoid; open = low-pressure forward-flush, closed = produce.

3. I/O Assignment

Channel numbers below are interpreted from the wiring diagram and should be confirmed against the final wiring.

Terminal	Signal	Type	Function
ADIO1	Pressure Meter IN	4-20 mA in (CIN)	Membrane feed pressure (high-pressure side)
ADIO2	Pressure Meter OUT	4-20 mA in (CIN)	Membrane concentrate (brine) pressure
ADIO3	Depth Meter - Clean Water Tank	4-20 mA in (CIN)	Product tank level
ADIO4	Depth Meter - Supply Tank	4-20 mA in (CIN)	Feed tank level
ADIO5	TDS Sensor IN	0-5 V in (AIN)	Feed water TDS
ADIO6	TDS Sensor OUT	0-5 V in (AIN)	Permeate TDS
ADIO7	Run/Stop selector	digital in (DIN)	Run enable / emergency stop / fault reset
ADIO8	10K NTC	NTC in (TIN)	RO pump motor temperature
DIO9	Flow Meter IN	pulse in (PCC)	Feed flow
DIO10	Flow Meter OUT	pulse in (PCC)	Permeate flow
DIO11	RGB LED - Red	output (POUT)	Status indicator
DIO12	RGB LED - Green	output (POUT)	Status indicator
DIO13	RGB LED - Blue	output (POUT)	Status indicator
DIO14	Flush valve	output (POUT, 24 V)	Brine-bypass / forward-flush valve
DIO15	RO Pump contactor	output (POUT, 24 V)	Pump motor ON/OFF
DIO16	Supply valve	output (POUT, 24 V)	Feed inlet valve

4. Sensor Scaling & Engineering Units

Sensor	Electrical	Engineering scale	Notes
Depth - both tanks	4-20 mA	4 mA = 0.00 m, 20 mA = 5.00 m (0-500 cm)	Linear
Pressure IN / OUT	4-20 mA	4 mA = 0 PSI, 20 mA = 300 PSI	Range extended to 300 PSI
TDS IN	0-5 V	0 V = 0, 5 V = 10,000 mg/L	Feed salinity
TDS OUT	0-5 V	0 V = 0, 5 V = 1,000 mg/L	Permeate quality
Flow IN / OUT	pulse	100 pulses = 1 L	Controller integrates to L/min
Motor temp	10K NTC	Standard 10K NTC (Beta 3435, 10 k ref)	Default characteristic

The pressure transmitters are now 0-300 PSI (20 mA = 300 PSI). This keeps the 150-200 PSI operating band comfortably mid-scale and, crucially, lets the controller read genuine over-200 PSI conditions for fouling detection (Section 8) and provides headroom for an over-pressure safety trip at 250 PSI.

5. Operating States

State	Condition	Pump	Valves	RGB
STOP (E-stop)	Run/Stop selector in STOP	OFF	Supply closed, Flush closed	OFF
STOPPED - Tank Full	Clean tank full	OFF	Closed	Blue solid
PAUSED - Supply Low	Supply tank below minimum	OFF	Closed	Blue blinking
STARTING	Permissives met; pressure settling	ON	Supply open, Flush closed	Green
RUNNING	Producing within spec	ON	Supply open, Flush closed	Green solid
FLUSHING	Periodic 5-min forward-flush	ON	Supply open, Flush open	Yellow solid
WARNING	Running but marginal	ON	Producing	Yellow flashing
FAULT	Latched error	OFF	Closed (safe)	Red

6. Control Logic & Sequence of Operation

6.1 Run/Stop selector (emergency stop & reset)

- A maintained Run/Stop selector is the master enable and emergency stop.
- STOP: immediately de-energises all outputs (pump OFF, both valves closed). Unconditional safe stop, bypasses minimum on-time.
- Moving to STOP also clears any latched fault.
- RUN: the controller runs the automatic sequence subject to the start permissives.

6.2 Power-on behaviour

- On power-up, if the selector is in RUN, the skid begins automatic operation immediately (no operator start action, no power-up flush).
- If the selector is in STOP at power-up, the skid stays halted until set to RUN.

6.3 Start permissives (all must be true to start the pump)

- Run/Stop selector = RUN; no latched fault.
- Supply tank level >= 50 cm.
- Clean-water tank level < restart threshold (380 cm).
- Motor temperature < 80 deg C.
- Minimum OFF-time (10 s) since the pump last stopped has elapsed.

6.4 Start sequence

1. Open the supply valve.
2. Start the RO pump (flush valve stays closed so the restrictor builds membrane pressure).
3. Settling window (25 s): production alarms (flow, recovery, pressure-band, TDS) are masked while pressure and flow stabilise. Safety items (motor temp, over-pressure, sensor fault, E-stop) remain active.
4. Enter RUNNING and begin evaluating all production monitors (Section 7).

6.5 Motor protection (anti short-cycle)

- Minimum ON-time = 10 s before a non-safety stop (tank full, supply low) is allowed.
- Minimum OFF-time = 10 s before the pump may restart.
- Safety stops (E-stop and faults) act immediately and ignore the minimum ON-time.

6.6 Periodic membrane flush

- After 60 minutes of continuous run time the controller performs a 5-minute forward-flush.
- During flush: pump ON, supply valve open, flush valve open (brine bypasses the restrictor) - low pressure, high cross-flow. RGB = yellow solid.
- On completion: close the flush valve, resume RUNNING and reset the run timer.

6.7 Normal stop - clean-water tank full (with hysteresis)

- At 400 cm, stop production (after the minimum ON-time). RGB = blue solid.
- Restart automatically when the level falls to 380 cm and permissives are met.

6.8 Supply-low pause (with hysteresis)

- Below 50 cm, stop the pump and wait. RGB = blue blinking.
- Resume automatically at 60 cm (and other permissives met).

7. Process Monitoring & Protection

Warnings and short-time trips use debounce timers to avoid nuisance reactions to transients.

Parameter	Normal band	Warning	Trip / action
Feed flow (IN)	~10 L/min (8-12)	Outside band	~0 with pump ON > 10 s -> dry-run FAULT
Recovery (permeate / feed)	10 - 30 %	Out of range	Sustained out of range -> WARNING
Permeate TDS (OUT)	< 200 mg/L	Rising toward 200	> 200 mg/L for 10 min -> FAULT stop
Pressure IN	150 - 200 PSI	> 200 (early fouling)	> 250 PSI -> over-pressure FAULT
Pressure OUT	150 - 200 PSI	Drift	> 250 PSI -> over-pressure FAULT
Motor temperature	< 80 deg C	Approaching 80	>= 80 deg C -> FAULT stop
Analog sensors (4-20 mA)	4 - 20 mA	-	< ~3.5 mA = broken wire -> sensor FAULT

8. Membrane Fouling Detection

- With the 0-300 PSI transmitters the controller can now directly detect feed pressure above 200 PSI.
- Cleaning-required indication: feed pressure > 200 PSI while permeate flow is low indicates a fouled/scaled membrane. This raises a maintenance warning (yellow fast blink).
- High feed pressure (> 200 PSI) with otherwise normal flow raises an early-fouling warning (yellow slow blink).

- Differential pressure (Pressure IN - Pressure OUT) can be trended as a secondary fouling indicator.
- An over-pressure above 250 PSI is treated as a safety FAULT (immediate stop).

9. Status Indication (RGB LED)

Only one state is shown at a time. Priority, highest to lowest: STOP (off) -> Fault (red) -> Flushing (yellow solid) -> Warning (yellow flashing) -> Running (green) -> Supply low (blue blink) -> Tank full (blue solid).

Indication	Meaning
Off	Run/Stop selector in STOP, or no power
Blue - solid	Stopped: clean-water tank full
Blue - blinking	Paused: supply tank low
Green - solid	Running normally / in spec
Yellow - solid	Flushing the membrane
Yellow - slow blink	Marginal: feed pressure high / low (early fouling)
Yellow - medium blink	Marginal: low/high flow or recovery out of band
Yellow - fast blink	Marginal: rising permeate TDS / membrane cleaning required
Red	Stopped on fault (see Section 10 blink codes)

10. Fault Handling & Reset

- On any fault: stop the pump immediately (bypassing minimum ON-time) and drive outputs to the safe state (pump OFF, both valves closed). The fault latches.
- Reset: move the Run/Stop selector to STOP (clears the latched fault), then back to RUN to resume; the minimum OFF-time still applies before restart.

Fault	Detection	Red code	Action
Motor over-temperature	NTC >= 80 deg C	1 blink	Latched stop
Dry-run / no feed flow	Flow IN ~0 with pump ON > 10 s	2 blinks	Latched stop
Permeate out of spec	TDS OUT > 200 mg/L for 10 min	3 blinks	Latched stop
Over-pressure / pressure fault	Pressure IN/OUT > 250 PSI	4 blinks	Latched stop
Sensor fault	4-20 mA input < ~3.5 mA (broken wire)	5 blinks	Latched stop

11. Default Parameters / Setpoints

Parameter	Value
Supply tank - minimum to run	50 cm
Supply tank - resume level	60 cm
Clean tank - full / stop	400 cm
Clean tank - restart level	380 cm
Operating pressure band (IN & OUT)	150 - 200 PSI
Pressure transmitter range	0 - 300 PSI (20 mA = 300 PSI)
Over-pressure safety trip	250 PSI
Permeate TDS limit	200 mg/L

Permeate TDS fault delay	10 min sustained
Feed flow target / band	~10 L/min (8 - 12)
Recovery target	10 - 30 %
Motor temperature limit	80 deg C (10K NTC)
Periodic flush interval	every 60 min of run time
Flush duration	5 min
Minimum pump ON-time	10 s
Minimum pump OFF-time	10 s
Start settling window	25 s
Dry-run grace time	10 s
Power-on	auto-run if Run/Stop = RUN

12. Out of Scope (this revision)

- MQTT / cloud connectivity, telemetry and remote dashboards.
- Data logging / historian.
- Temperature compensation of flow / TDS.
- PID or any closed-loop modulating control (pump is simple ON/OFF).
- Start-up / power-up flush and permeate flush-to-drain at start.
- Feed / booster pump (supply is gravity-fed).

13. Open Points to Confirm

5. Confirm the feed-flow target/band (~10 L/min, 8-12) and the meter K-factor (100 pulses/L).
6. Confirm the interpreted ADIO/DIO channel numbers against the final wiring.
7. Confirm that out-of-spec permeate (TDS > 200 for 10 min) should STOP the skid (no divert-to-drain valve fitted).
8. Confirm warning thresholds, yellow flash-rate mapping, and debounce times.
9. Confirm the 250 PSI over-pressure safety trip value.
10. Should the fouling 'cleaning required' state only warn, or escalate to a stop after a set time?